

Cucker-Smale model with singular communication weights

Shih-Hsien Fest on Hyperbolic Conservation Laws and Kinetic Theory

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1. Introduction
2. Kinetic Cucker-Smale model
3. Hydrodynamic Cucker-Smale model
4. Summary

Introduction

Examples for flocking



(a) P. by Alastair Rae



(b) P. by Bruno de Giusti



(c) P. by Marco Bianchetti

- In [Cucker-Smale, '07], a model about mathematical description for flocking was presented:

$$\begin{cases} \dot{x}_i = v_i, & t > 0, \quad 1 \leq i \leq N, \\ \dot{v}_i = \frac{1}{N} \sum_{j=1}^N \phi(x_i - x_j)(v_j - v_i), \\ (x_i(0), v_i(0)) = (x_i^0, v_i^0), \end{cases} \quad (\text{CS})$$

where $\phi(x) = \bar{\phi}(|x|)$ and $\bar{\phi} : [0, \infty) \rightarrow \mathbb{R}_+$ is a **communication weight function** which is bounded, monotonically decreasing and Lipschitz continuous.

Kinetic description of the CS system

Via the mean-field limit $N \rightarrow \infty$ in (CS), we can find the one-particle distribution function $f = f(t, x, v)$ the kinetic density satisfying the **kinetic Cucker-Smale** model [Ha-Liu, '09], a.k.a. Vlasov-alignment model:

$$\begin{cases} \partial_t f + v \cdot \nabla_x f + \nabla_v \cdot (F[f]f) = 0, & x, v \in \mathbb{R}^d, \ t > 0, \\ F[f](t, x, v) = (\phi \star m)(t, x) - v(\phi \star \rho)(t, x), \\ f(0, x, v) = f_0(x, v), \end{cases} \quad (\text{KCS})$$

where

$$\rho(t, x) := \int_{\mathbb{R}^d} f(t, x, v) \, dv, \quad m(t, x) := \int_{\mathbb{R}^d} v f(t, x, v) \, dv.$$

Hydrodynamic Cucker-Smale model

To derive a hydrodynamic model from (KCS), we introduce

$$\begin{aligned}\rho(t, x) &:= \int_{\mathbb{R}^d} f dv, & (\rho u)(t, x) &:= \int_{\mathbb{R}^d} v f dv, \\ (\rho E)(x, t) &:= \frac{1}{2} \rho |u|^2 + \rho e, & \rho e &:= \frac{1}{2} \int_{\mathbb{R}^d} |v - u(x, t)|^2 f dv,\end{aligned}$$

and one introduces a **mono-kinetic** ansatz for f as a closure condition:

$$f(t, x, v) = \rho(t, x) \delta_{(v - u(x, t))}(v).$$

Then, we multiply 1 and v to (KCS), and integrate with respect to v to obtain **hydrodynamic Cucker-Smale model** model, a pressureless Euler system with flocking dissipation:

$$\left\{ \begin{array}{l} \partial_t \rho + \nabla \cdot (\rho u) = 0, \quad t > 0, \quad x \in \mathbb{R}^d, \\ \partial_t (\rho u) + \nabla \cdot (\rho u \otimes u) \\ \quad = \rho \int_{\mathbb{R}^d} \phi(x-y)(u(t,y) - u(t,x)) \rho(t,y) dy, \\ (\rho(0, x), u(0, x)) = (\rho_0(x), u_0(x)), \quad x \in \mathbb{R}^d. \end{array} \right. \quad (\text{HCS})$$

(See [\[Figalli-Kang, '19\]](#) for its rigorous derivation)

- Particle model
 - Emergence of mono-cluster or multi-cluster flocking
[Cucker-Smale, '07; Ha-Tadmor, '08; Ha-Liu, '09; Cho et al, '16]
 - Effects of white noises
[Cucker-Mordecki, '08; Ha-Lee-Levy, '09; Ahn et al, '10]
 - Time-delay effects
[Pignotti-Vallejo, '17; Choi-Haskovec, '17; Choi-Li, '18]
 - Application to flight formation [Perea et al, '09]
 - Collision avoidance [Ahn et al, '12, Carrillo et al, '17]
 - Generalized network structures including hierarchical leadership
[Shen, '07; Li-Xue, '10; Cucker-Dong, '16; Dong-Qiu, '17]
 - Mean-field limit
[Ha-Liu, '09; Carrillo et al, '14; Ha et al, '18]

- Kinetic description
 - Well-posedness of strong/weak solutions with regular kernels
[Ha-Tadmor, '08; Carrillo et al, '10]
 - Global solutions with non-compact velocity support
[Chen-Yin, '19; Wang-Xue, '22]
 - Hydrodynamic limits
[Karper et al, '15; Figalli-Kang, '18; Choi-Kim, '23]
- Hydrodynamic description
 - Well-posedness of solutions with regular kernels
[Ha et al, '15; Amadori-Christoforou, '22]
 - Well-posedness of strong solutions with hypersingular kernels
[Do et al, '17; Danchin et al, '18; Shvydkoy-Tadmor, '18; Tan, '20]

- Our interest lies in the case of **singular** communication weight, i.e. $\phi(x) = |x|^{-\gamma}$, $\gamma > 0$.
- When the weight is singular, the problem of being well-posedness for particle, kinetic, and hydrodynamic equations arises.
- For particle model (CS), one attains the collision avoidance for generic initial data when $\gamma \geq 1$ [Carrillo et al, '17].
- When $\gamma < 1$, system (CS) may experience finite-in-time collision [Peszek, '15].

Kinetic Cucker-Smale model

Well-posedness of kinetic Cucker-Smale model

- For kinetic equations (KCS), the method of characteristics gives the existence and uniqueness of weak [Carrillo et al, '14] and classical solutions when $\gamma < d - 1$, since $\nabla\phi$ is integrable near origin.
- However, when $\gamma \geq d - 1$, $\nabla\phi$ is **not** integrable near origin and hence usual estimates may not directly work.
- So, we investigate the well-posedness of strong and weak solutions for $\gamma \geq d - 1$.

Preliminaries

For the uniqueness of weak solutions, we introduce the p th-order Wasserstein distance:

$$d_p(\rho_1, \rho_2) := \begin{cases} \left(\inf_{\gamma} \iint_{\mathbb{R}^k \times \mathbb{R}^k} |x - y|^p d\gamma(x, y) \right)^{\frac{1}{p}} & \text{if } p \in [1, \infty), \\ \inf_{\gamma} \left(\sup\{|x - y| : x \in \text{supp } \rho_1, y \in \text{supp } \rho_2\} \right) & \text{if } p = \infty, \end{cases}$$

for $\rho_1, \rho_2 \in \mathcal{P}_p(\mathbb{R}^d)$, $p \in [1, \infty)$, where γ is any transference plan between ρ_1 and ρ_2 , i.e. for any $\psi \in \mathcal{C}_b(\mathbb{R}^k)$,

$$\begin{aligned} \iint_{\mathbb{R}^d \times \mathbb{R}^d} \psi(x) d\gamma(x, y) &= \int_{\mathbb{R}^d} \psi(x) \rho_1(x) dx, \quad \text{and} \\ \iint_{\mathbb{R}^d \times \mathbb{R}^d} \psi(y) d\gamma(x, y) &= \int_{\mathbb{R}^d} \psi(y) \rho_2(y) dy. \end{aligned}$$

Here $\mathcal{P}_p(\mathbb{R}^k)$ stands for the set of probability measures with bounded p th-order moment.

Main result 1

Theorem (Choi-J., '24)

Let $d \geq 1$ and $\gamma \in (0, d)$. Suppose that the initial data f_0 is compactly supported in velocity and satisfies

$$f_0 \in L^1_+ \cap L^\infty(\mathbb{R}^d \times \mathbb{R}^d) \quad \text{and} \quad |x|f_0 \in L^1(\mathbb{R}^d \times \mathbb{R}^d).$$

Then there exist $T > 0$ and at least one weak solution f to (KCS).

Moreover, if $\gamma \in (0, d - 1]$, the solution obtained above exists uniquely.

Step 1: Existence

We first regularize the equation (KCS) as follows:

$$\partial_t f^\varepsilon + v \cdot \nabla_x f^\varepsilon + \nabla_v \cdot [(\phi^\varepsilon \star m^\varepsilon - v \phi^\varepsilon \star \rho^\varepsilon) f^\varepsilon] = 0, \quad (\varepsilon\text{-KCS})$$

subject to the initial data:

$$f^\varepsilon(0, x, v) = (f_0 \star_{x,v} \theta_\varepsilon)(x, v), \quad (x, v) \in \mathbb{R}^d \times \mathbb{R}^d,$$

where $\theta_\varepsilon(x, v) = \varepsilon^{-2d} \theta(x/\varepsilon, v/\varepsilon)$ is the standard mollifier and $\phi^\varepsilon = \phi^\varepsilon(x)$ is written as

$$\phi^\varepsilon(x) := \frac{1}{(\varepsilon + |x|^2)^{\frac{\gamma}{2}}}.$$

By obtaining uniform-in- ε estimates for $\|f^\varepsilon(t)\|_{L^1 \cap L^\infty}$ for short time and using a variant of velocity averaging lemma [Karper et al, '13], one gets the existence.

Step 2: Uniqueness

- For each $i = 1, 2$, we consider the solution f_i corresponding to the same initial data f^{in} , and their characteristics $Z_i(t; s, z) := (X_i(t; s, z), V_i(t; s, z))$, $z := (x, v)$:

$$\begin{aligned}\frac{d}{dt} X_i(t; s, z) &= V_i(t; s, z), \quad t, s \in (0, T), \\ \frac{d}{dt} V_i(t; s, z) &= (\phi \star m_i)(X_i(t; s, z)) - V_i(t; s, z)(\phi \star \rho_i)(X_i(t; s, z))\end{aligned}$$

subject to $Z_i(s; s, z) = z$.

- Then we know that

$$d_1(f_1(t), f_2(t)) \leq \int_{\mathbb{R}^d \times \mathbb{R}^d} |Z_1(t; 0, z) - Z_2(t; 0, z)| f^{in}(z) dz =: Q(t),$$

and use the log-Lipschitz property of $\phi \star m_i$ and $\phi \star \rho_i$ to yield

$$\left. \frac{d}{dt} Q(t) \right|_{t=t_0^+} \leq C d_1(f_1(t_0), f_2(t_0)) (1 - \log^-(d_1(f_1(t_0), f_2(t_0)))) ,$$

which completes the proof.

Main result 2

We define the following weighted L^2 -Sobolev norm:

$$\|f\|_{H_{x,v}^{s,2}}^2 := \int_{\mathbb{R}^d \times \mathbb{R}^d} \left(|\langle x \rangle^2 f|^2 + |\Lambda_x^s f|^2 + |\Lambda_v^s f|^2 \right) dx dv,$$

where $\langle x \rangle = (1 + |x|^2)^{1/2}$ and $\Lambda^s := (-\Delta)^{\frac{s}{2}}$.

Theorem (Choi-J., '23)

Suppose that $\gamma \in (d-1, d + \frac{1}{4}) \setminus \{d\}$. Then for any $s > \frac{d}{2} + 1$ and the initial data $f_0 \in (L_+^1 \cap H_{x,v}^{s,2})(\mathbb{R}^d \times \mathbb{R}^d)$ compactly supported in v , there exists $T > 0$ such that a unique solution f to (KCS) corresponding to the initial datum f_0 exists in $L^\infty([0, T]; (L_+^1 \cap H_{x,v}^{s,2})(\mathbb{R}^d \times \mathbb{R}^d))$ and it has a compact support in v .

Here, we separately consider two cases (i) $\gamma \in (d-1, d)$ and (ii) $\gamma \in (d, d+1/4)$.

(i) ($\gamma \in (d-1, d)$): We set $\alpha = d - \gamma \in (0, 1)$. Then, F can be rewritten as

$$F[f] := \Lambda^{-\alpha}[m] - v\Lambda^{-\alpha}[\rho],$$

where

$$\rho := \int_{\mathbb{R}^d} f \, dv, \quad m := \int_{\mathbb{R}^d} vf \, dv,$$

and $\Lambda^{-\alpha} := (-\Delta)^{-\alpha/2}$ is the Riesz operator. In this case, we set

$$u_1(x) := \Lambda^{-\alpha}[m], \quad u_2(x) := \Lambda^{-\alpha}[\rho].$$

(ii) ($\gamma \in (d, d + \frac{1}{4})$): We write $\beta = \gamma - d \in (0, \frac{1}{4})$. Then note that

$$\frac{1}{|x|^{d+\beta}} = \nabla_x \cdot \left(\frac{x}{-\beta|x|^{d+\beta}} \right),$$

and this gives

$$\begin{aligned} F[f] &= \nabla_x \cdot \left[\int_{\mathbb{R}^d \times \mathbb{R}^d} \frac{(x-y)}{-\beta|x-y|^{d+\beta}} \otimes (w-v) f(y, w) dy dw \right] \\ &= C(d, \gamma) \left((\nabla_x \cdot \Lambda^{-2+\beta}[\nabla m]) - v(\nabla_x \cdot \Lambda^{-2+\beta}[\nabla \rho]) \right). \end{aligned}$$

In this case, we set

$$u_1(x) := C(d, \gamma) \nabla \cdot \Lambda^{-2+\beta}[\nabla m], \quad u_2(x) := C(d, \gamma) \nabla \cdot \Lambda^{-2+\beta}[\nabla \rho].$$

- In either case, (KCS) can be reformulated as

$$\partial_t f + v \cdot \nabla_x f + \nabla_v \cdot [(u_1 - v u_2) f] = 0. \quad (1)$$

- For the case (i), thanks to the compact support in v , we have $\|u_1\|_{H_x^s} \sim \|m\|_{H^{s-\alpha}}$ and $\|u_2\|_{H_x^s} \sim \|\rho\|_{H^{s-\alpha}}$ with $\alpha = d - \gamma$. Thus, we can deal with the first case (i) rather readily.
- On the other hand, in the case of **hypersingular** communication weights, i.e. $\gamma \in (d, d + 1/4)$, we formally expect $\|u_1\|_{H_x^s} \sim \|m\|_{H^{s+\beta}}$ and $\|u_2\|_{H_x^s} \sim \|\rho\|_{H^{s+\beta}}$ with $\beta = \gamma - d \in (0, \frac{1}{4})$.
- Thus, we need to gain additional regularity for ρ and m compared to the solution f . For this, we employ the velocity averaging lemma [Glassey, '96].

Hydrodynamic Cucker-Smale model

Hydrodynamic Cucker-Smale model

- Recall that the system reads

$$\begin{cases} \partial_t \rho + \nabla \cdot (\rho u) = 0, & t > 0, \quad x \in \mathbb{R}^d, \\ \partial_t u + u \cdot \nabla u = \int_{\mathbb{R}^d} \phi(x-y)(u(t,y) - u(t,x))\rho(t,y)dy. \end{cases} \quad (\text{HCS})$$

- When the kernel is hypersingular, i.e. $\gamma \in (d, d+2)$, has been mainly considered, where one may rewrite the alignment force as

$$-\rho(-\Delta)^{\frac{\sigma}{2}} u + \int_{\mathbb{R}^d} \frac{(u(y) - u(x))(\rho(y) - \rho(x))}{|x - y|^{d+\sigma}} dy, \quad \sigma =: \gamma - d \in (0, 2),$$

- This gives elliptic regularity to the model and its global well-posedness has been investigated [Do et al, '18; Danchin et al, '19; Lear et al, '22].
- However, when $\gamma < d$, the one-dimensional case has only been discussed [Shvydkoy-Tadmor, '18; Tan, '20].

- Without loss of generality, we may assume

$$m_0(t) =: \int_{\mathbb{R}^d} \rho(t) \, dx = 1, \quad x_c(t) =: \int_{\mathbb{R}^d} x \rho(t) \, dx = 0 \quad \text{and}$$

$$m_c(t) =: \int_{\mathbb{R}^d} (\rho u)(t) \, dx = 0 \quad \text{for all } t \geq 0.$$

- We also define diameter functions as

$$D(t) := \sup\{|x - y| : x, y \in \text{supp } \rho(t, \cdot)\},$$

$$V(t) := \sup\{|u(t, x) - u(t, y)| : x, y \in \text{supp } \rho(t, \cdot)\}.$$

We say (ρ, u) is a solution to (HCS) on $[0, T)$ corresponding to the initial data (ρ_0, u_0) if the pair satisfies the following:

1. $(\rho, u) \in L^\infty([0, T]; L^1_+ \cap L^\infty(\mathbb{R}^d)) \times L^\infty([0, T]; W^{1,\infty}(\bar{B}(0, R)))$ for some $R > 0$.
2. (ρ, u) satisfies (HCS) in the sense of distributions.

Theorem (Choi-J., '24)

Suppose that the initial data (ρ_0, u_0) satisfy

- (i) ρ_0 is compactly supported and $\rho_0 \in L^\infty(\mathbb{R}^d)$,
 - (ii) $V(0) < \int_{D(0)}^\infty \phi(s) ds$, and
 - (iii) $\|u_0\|_{W^{1,\infty}(\bar{B}(0,2D^\infty))} < \varepsilon_0$ for D^∞ and sufficiently small $\varepsilon_0 \ll 1$,
- where D^∞ is defined as

$$D^\infty := \begin{cases} (D(0)^{1-\gamma} + (1-\gamma)V(0))^{\frac{1}{1-\gamma}} & \text{if } \gamma \neq 1 \\ D(0)e^{V(0)} & \text{if } \gamma = 1 \end{cases}.$$

Then there exists a unique global-in-time solution (ρ, u) to the initial value problem (HCS) with $R = 2D^\infty$.

Strategy of the proof

- Obtain the flocking estimate $V(t) \leq V(0)e^{-\phi(D^\infty)t}$ based on the method of characteristics and extend this to a bit larger area; we set

$$V^\infty(t) := \sup\{|u(t, x) - u(t, y)| : x, y \in \bar{B}(0, 2D^\infty)\}.$$

Then one gets

$$V^\infty(t) \leq V^\infty(0)e^{-\phi(3D^\infty)t}.$$

- Based on a priori assumption

$$\sup_{0 \leq t \leq T} \|u(t)\|_{W^{1,\infty}(\bar{B}(0, 2D^\infty))} < \varepsilon_1 \ll 1,$$

we can obtain

$$\sup_{0 \leq t \leq T} \|\rho(t)\|_{L^\infty} \leq C_1 \|\rho_0\|_{L^\infty},$$

$$\sup_{0 \leq t \leq T} \|u(t)\|_{W^{1,\infty}(\bar{B}(0, 2D^\infty))} \leq C_2 \|u_0\|_{W^{1,\infty}(\bar{B}(0, 2D^\infty))}$$

with C_i 's independent of T and this implies the global existence.

- For uniqueness, consider two solutions (ρ_i, u_i) ($i = 1, 2$) corresponding to the same initial data (ρ_0, u_0) and their forward characteristics:

$$\partial_t \eta_i(t, x) = u_i(t, \eta_i(t, x)), \quad \eta_i(0, x) = x \in \text{supp} \rho_0, \quad i = 1, 2.$$

Then we have

$$d_\infty(\rho_1(t), \rho_2(t)) \leq \sup_{x \in \text{supp} \rho_0} |\eta_1(t, x) - \eta_2(t, x)|,$$

which implies

$$d_\infty(\rho_1(t), \rho_2(t)) \leq C \int_0^t \|(u_1 - u_2)(s)\|_{L^\infty(\bar{B}(0, 2D^\infty))} ds.$$

- To estimate $u_1 - u_2$, we write $\mathcal{K}(x, y) := \phi(x - y)(u_2(y) - u_2(x))\chi(y)$, where $\chi \in C_c^\infty(\mathbb{R}^d)$ satisfies

$$\mathbf{1}_{B(0, D^\infty)} \leq \chi \leq \mathbf{1}_{B(0, 3D^\infty/2)}.$$

Then almost the same argument as in [\[Loeper, '06\]](#) gives

$$\left\| \int_{\mathbb{R}^d} \mathcal{K}(\cdot, y)(\rho_1 - \rho_2)(y) dy \right\|_{L^\infty(\bar{B}(0, 2D^\infty))} \leq \max\{\|\rho_1\|_{L^\infty}, \|\rho_2\|_{L^\infty}\}^{1/2} d_\infty(\rho_1, \rho_2).$$

and this imply

$$\|(u_1 - u_2)(t)\|_{L^\infty(\bar{B}(0, 2D^\infty))} \leq C \int_0^t d_\infty(\rho_1(s), \rho_2(s)) ds.$$

Summary

- Kinetic Cucker-Smale model with singular communication weights
 - Uniform bounds w.r.t. regularization parameter together with velocity averaging lemma give the existence of weak solutions.
 - For uniqueness, we can employ 1-Wasserstein metric estimates thanks to log-Lipschitz property for interaction forces.
 - To establish the local well-posedness of strong solutions, we view the communication weight function as the Fourier multiplier.
- Hydrodynamic Cucker-Smale model with singular communication weights
 - We use the flocking estimates to obtain the uniform-in-time bounds for the solution, and hence the global existence.
 - For the uniqueness, we employed ∞ -Wasserstein metric for the densities and L^∞ -norm for velocities.

Future works & open questions

- Particle model
 - Collision avoidance of discrete model for generic non-collisional initial data
- Kinetic model
 - Global existence of strong solutions
 - Strong solutions with non-compact v -supports
 - Ill-posedness of solutions
- Hydrodynamic model
 - Global solutions without compact support condition on density when $\gamma < d$
 - Well-posedness of solutions in Hölder spaces when $\gamma > d$

THANK YOU FOR YOUR ATTENTION.